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Characterizing Macropores in Soil by Computed Tomography

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ABSTRACT

Cross sections of a 203-mm diam. soil core taken from a site under Kentucky bluegrass (*Poa pratensis L.*) were scanned using x-ray computed tomography (CT) to determine if macropores such as cracks, earthworm holes and root channels could be distinguished and characterized. The core was physically sectioned at the same locations as the CT scans to verify the size and location of pores seen on the CT images. Macropores 1 mm and larger could be easily and quickly distinguished by the method. By manipulation of the scanner images, air-filled pores, roots and stones were identified within the soil matrix. A wet bulk density plus an air-filled macroporosity were calculated for scans. Prior to sectioning, a dye solution was ponded on the surface of the core and allowed to infiltrate to assess the continuity of pores through the core.

SOIL MACROPORES and their associated flow and transport phenomena have long been known to exist as indicated by Thomas and Phillips (1979) and Beven and Germann (1982), but only recently has much emphasis been placed on their importance in hydrologic processes. As Beven and Germann (1982) explained, macroporosity is difficult to define because it is a relative term. For the purpose of this paper we are using the term macropores to mean pores that are significantly larger than the intergranular or interaggregate pores. Macropores may be produced by either physical or biological processes. Examples of physical processes include the shrink/swell phenomena caused by the drying and wetting of certain soils, hydraulic pressures causing subsurface erosion, freezing and thawing, and man-induced processes such as tillage. Biological processes include actions by earthworms, insects and other animals and those caused by plant roots. Beven and Germann (1982) provide a thorough overview of the different types of macropores.

The presence of macropores in soils has been shown to greatly influence the infiltration rate and movement of water within a soil profile (Edwards et al., 1979; Germann and Beven, 1981; Germann and Beven 1985; and Lee, 1985). Water penetration into a homogeneous soil system without macropores occurs through the small pores or voids among the grains or aggregates, and usually the flow is considered to follow Darcy's law. For this type of flow (matrix flow), the flow capacity of the soil is governed by the size and connectivity of the intergranular or interaggregate soil voids. However, where macropores exist in a soil, water flow in the soil may be dominated by flow through the macropore system, especially for near-saturated conditions at the surface.

Flow through macropores is much faster than through the soil matrix system and is usually non-Darcian (Edwards et al., 1979; Germann and Beven, 1985). Rapid flow has major implications for groundwater quality because nutrients or chemicals applied at the surface may be quickly delivered to the underlying groundwater system without passing through the soil matrix where they could be adsorbed and/or used by plants. For instance, in recent field and computer studies, Steenhuis et al. (1986) and Richard and Steenhuis (1987) demonstrated that movement of surface water and solutes to drainage tile lines was more than two orders of magnitude faster than would be predicted by Darcian flow through the soil matrix. Other implications of the presence or absence of macropores may include the surface runoff rate, aeration of the soil and penetration of root systems into dense soil layers.

To assess the impact of macropores on the flow and transport properties of soils, the size, number, type, distribution and continuity of macropores in the soils must be characterized. This characterization can be accomplished in two ways. The first approach is to directly measure the pores at different depths in a soil profile. Included in this approach are: (i) analysis of

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photographs of thin sections of small soil cores, after impregnation with hardeners and dye tracers (Bouma et al., 1977; Beven and Germann, 1982; Ringrose-Voase and Bullock, 1984); (ii) field measurement using photographs as described by Edwards et al. (1988), and Smettem and Collis-George (1985); and (iii) scanning techniques using computed tomography.

The second approach is to indirectly characterize macropore structure by measuring water flow and/or chemical transport. This approach has been used by Watson and Luxmoore (1986) and Moore et al. (1986) using suction infiltrometers for measurement of water flow in forested soils. Anderson and Bouma (1977), Bouma and Wosten (1979) and others have used chloride breakthrough curves from undisturbed soil columns as another indirect means of characterizing macropores.

The objective of the present study was to evaluate the use of computed tomography (CT), also known as computer aided tomography (CAT), to directly and nondestructively characterize macropore structure in soil columns. The CT scanner was developed for medical purposes but has proven useful in the field of soil science for determination of soil water content and bulk density (Petrovic et al., 1982; Hainsworth and Aylmore, 1983; Crestana et al., 1985; Hainsworth and Aylmore, 1986; Anderson et al. 1988; and Tollner et al., 1987).

MATERIALS AND METHODS

Soil Core

An undisturbed core, approximately 203 mm in diameter, was excavated from a research plot at the Univ. of Minnesota in St. Paul, MN. The core was obtained from an uncultivated field border that had been vegetated with Kentucky bluegrass for many years. The only cultural practice was periodic mowing of the grass. The core was taken in June during a period of heavy rains in the area. Soil moisture was near field capacity. The soil from which the core was taken was a Waukegan silt loam (fine-silty over sandy or sandy skeletal, mixed, mesic, Typic, Hapludoll). These soils are characteristically well drained with moderate permeability and have a weak fine subangular blocky structure in their top horizons. Field slopes at the site are less than two percent.

The core was obtained by driving a PVC pipe (nominal inside diameter of 203 mm) into the soil in increments of about 100 mm and then digging around the outside of the pipe. This process was repeated until a total depth of about 700 mm was reached, after which the column was cut at its base and removed from the hole. Melted paraffin was poured around the top inside circumference of the core to seal off any flow along the core walls during dye tracing experiments. The ends of the PVC pipe were sealed with plastic to prevent loss of material during the scanning process.

In addition to the undisturbed field core, a soil column was prepared with artificially produced macropores of known size and location. This soil column was considered to provide controlled conditions for testing the accuracy of the CT scanning method in detecting macropores. The column consisted of a 105-mm long section of the PVC pipe packed with the Waukegan soil to a dry bulk density of 1.45 MG m^{-3}. Holes varying in diameter from about 1.6 mm to about 8 mm were formed longitudinally along the column by inserting wire probes or by the inclusion of plastic or glass tubing. The plastic and glass tubing was used to form the larger pores. These tubes were left in the column during the CT scans.

CT Scan Procedure

The CT scanner used in this study was a Seimens SOMATOM DR3 using version F2 software located at the Univ. of Minnesota Hospital. In the CT scanning process an x-ray source and an array of detectors rotate about the object producing multiple sets of data from projections of an x-ray beam through an object. Data from several hundred projections are reconstructed into an image.

The CT scan image is composed of numerous picture elements (pixels), each of which corresponds to a position in the object being scanned. The attenuation of the x-ray beam through each position of the object is directly proportional to the density and atomic composition of the material in that position. A brightness or intensity value is assigned to each pixel of the image, based on the x-ray attenuation in that pixel. The image reconstruction process has been described by Brooks and DeChiro (1976) and reviewed in several articles regarding applications to soil water content and bulk density determinations (Anderson et al. 1988; Hainsworth and Aylmore, 1983).

The scan system parameters were set to 125 peak kV, 700 mA-s and 14-s scan time to provide high detail, low noise projections. Attenuation values for each pixel were calculated from the projection data using a high resolution, edge enhancing filter that is proprietary to the system. Each image was obtained using 1440 projections (.025° projection^{-1}). The field of view, i.e. the cross sectional dimensions, for the 203-mm diam. core was 250 mm by 250 mm giving a pixel size of approximately 0.536 mm by 0.536 mm. The x-ray beam width or "slice" thickness of each image was 8 mm for the packed soil column, and 2 mm for the field core. The distance between scans was 10 mm for the top 100 mm of the field core, 20 mm for the middle portion (100–300 mm) of the core and 50 mm for the lower part of the core. Images of scans were obtained on photographic film.

Sectioning of the Core

After the CT scans were obtained, water was ponded on the surface and allowed to infiltrate into the field core at several heads, both positive and negative, using a disc infiltrometer similar to that described by Nieber and Walter (1981). In addition, a solution of methylene blue dye was allowed to infiltrate into the core under ponded conditions to identify flow paths through the core and macropore connectivity to the soil surface. The outside of the PVC pipe was marked at the location of each scan and the "front" or topside of the pipe in its horizontal scanning position was marked so the core could be sectioned at the location of the scan images. The core was sectioned about 1 wk after ponding with dye. Consequently, the core was near field capacity in water content when sectioned.

The first step in sectioning the core was to saw through the PVC pipe with a hack saw at marked scan locations. Various methods for cutting the soil were evaluated. These included the use of hand and power saws, fine wire, sharp knives, and sawing after freezing the core. All these methods smeared the soil making detection of macropores difficult, however. The best method was to break or manually shear the soil at the desired cross section. Manual shearing did not smear the soil but had the disadvantage of at times producing an uneven break along the desired plane. When uneven breaks occurred, a straight pin and a chisel were used to produce a clean break in the vicinity of the scan plane. A shop vacuum cleaner was used to remove loose particles on the soil surface that might cover macropores.

After obtaining a fairly even surface, color-coded pins were stuck into the surface of each section to identify holes, cracks, roots, stones and filled holes (plugs). Stained features such as holes and roots were differentiated from unstained features. Photographs of each section were taken with and without the pins. The sections were visually compared with the

CT scan images at those sections to determine if the macropores shown on the images coincided with those in the sections. As an additional verification of the accuracy of the scan process, the images of the packed soil column were checked for number, size and location of macropores.

The connectivity of various macropores to the surface was assessed by the presence of dye in successive photographs of sections. In addition, after dividing the core into two equal lengths, a very fine sand was poured into two large macropores, about 7 mm in diameter, appearing at the ends of each section to assess the continuity of those macropores.

Data Analysis

Reconstructed images can be displayed on a black and white, CT-dedicated monitor and manipulated to accentuate features by changing the "window" and/or center "brightness" values used for different display shades. The window and center values are in Hounsfield units which are obtained by subtracting the attenuation value of water from the attenuation value for the pixel, dividing by the value of water and multiplying by 1000 (Brooks and DiChiro, 1983). The Hounsfield value for water is therefore zero. The window is the range of values covered by the shades of gray, while the center value is the median value for the range.

Values for specific points on the image can be obtained at the monitor by use of a screen pointer or wand. However, a distribution of values for exact locations or the entire scan is difficult to obtain from the CT dedicated monitor. In addition access to the CT dedicated monitor is limited due to the priority for medical use. To overcome these problems, data for each scan were recorded on magnetic tape, then transferred to a microcomputer. Programs were developed whereby images could be displayed using either black and white or color-coded ranges of attenuation values. Different features in the scans could be accentuated by this method and portions of an image enlarged for greater detail.

The availability of the images on the microcomputer allowed statistical analyses to be performed on all or parts of individual scans and related to values for known materials. By comparing attenuation coefficients with the known value for water, a wet bulk density for each scan was calculated. This approach assumes a linear relationship between attenuation value and electron density and between electron density and mass, which are good assumptions for the materials and energy levels employed here (Morgan, 1983; Anderson et al., 1988). It should be noted that these are total densities and not dry bulk densities for the soil. The linear relationship is based on an attenuation value of -1000 Hounsfield units for a density of zero and a Hounsfield unit of zero for a density of 1.00 Mg m^{-3} (water).

The air-filled porosity of macropores was also calculated for each scan by summing the number of pixels having values less than the maximum attenuation value for an air-filled pore. The maximum value for air in a pore was obtained using values from the artificial "macropores" in the packed soil column and from large (approximately 7-mm diam.) pores found in the field core images and verified in the physical sections. There was actually a range of values for air as determined from the scans. The value is dependent

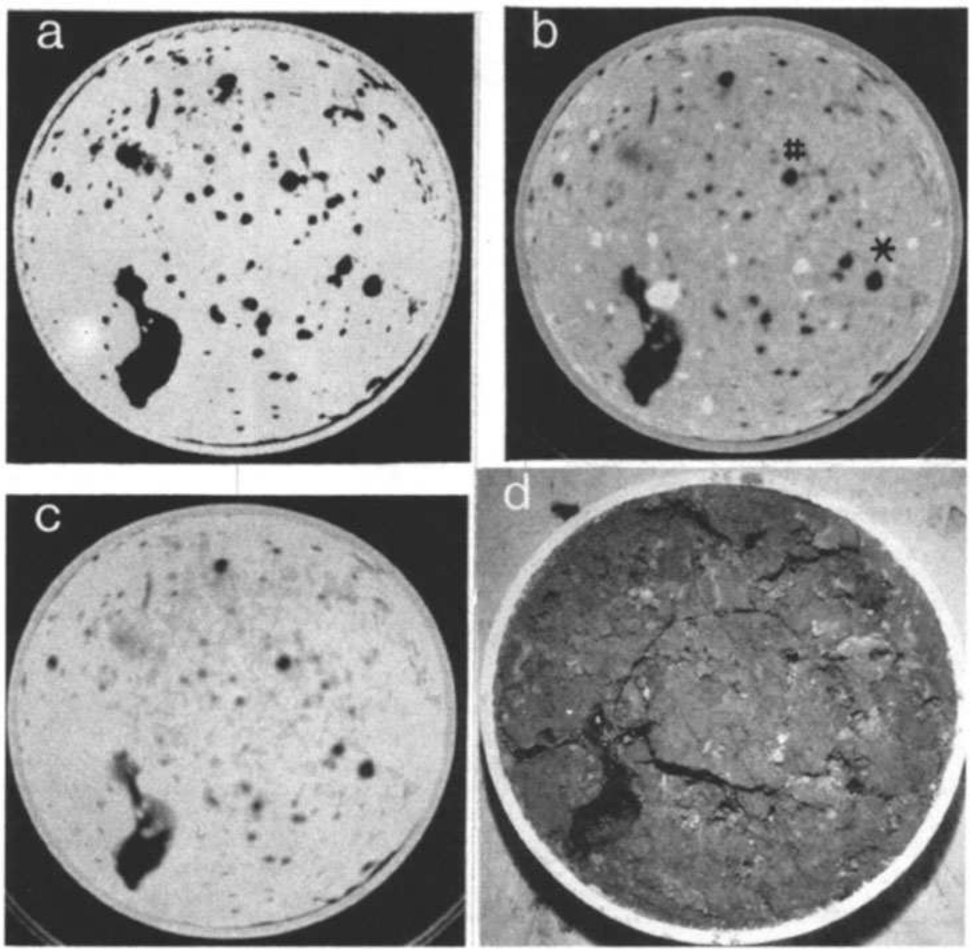

Fig. 1. CT images and section photo of field core. Depth = 290 mm. Values in Hounsfield units. a. Image with center value of 800 and window of 200. b. Image with center value of 600 and window of 1600. c. Image with center value of zero and window of 2000. d. Photo of physical section.

on the size of pore due to the reconstruction algorithm and geometry of the scanner. A Hounsfield value of −24 was used as the maximum value for an air-filled pore for the purposes of this study. It was reasoned that no material or combination of materials other than air present in the pixel should have a lesser value. Separate experiments not reported in this paper where air, roots, and water were scanned within a packed soil column conformed that this value is reasonable for use in detecting air-filled pores.

A size distribution of macropores for the images was obtained using color-coded displays on a microcomputer. The size for each pore was determined by counting the number of contiguous pixels having an attentuation value corresponding to that of air. Since the area represented by one pixel was known, the actual size of individual pores could be calculated. This distribution was done for each scan of the field core, and shows how the number of different sized pores changes with the depth for the core.

RESULTS

Images reconstructed using three different window and center values on a microcomputer plus the associated photo of the physical section are shown in Fig. 1 at a depth of 290 mm in the core. Black areas indicate material with a low attenuation value (low density), while white areas indicate material with a high attentuation value (high density). The image in Fig. 1a was reconstructed using a high center value (800) and a small window (200). Consequently, there are large amounts of both black and white areas but small amounts of gray areas. The black areas could correspond to water, air or organic matter, while the white areas are either soil matrix or stones. In Fig. 1b the center value is set at 600 and the window at 1600, resulting in black areas that are air, light areas that are soil and white areas that are small stones. In Fig. 1c the center value is set at zero and the window at 2000, resulting in more detail of the density differences within the soil matrix but less detail for distinguishing the air-filled macropores. In general the values used in generating Fig. 1b produce the best image in comparison with features shown in the photo of the section, Fig. 1d. The cracks in Fig. 1d are not seen in the corresponding images and are thought to have been created during the sectioning process. The large dark area in the photo and images is an ant cavity.

Images reconstructed on the microcomputer are presented in Fig. 2 for various scans of the field core. In these figures the center and window values are set as in Fig. 1b. The gray areas in all images require further interpretation from that given above and will be discussed later. The symbols * and # on the images indicate the continuous pores into which the fine sand was poured. The sand passed completely through these two large (approximately 7 mm diameter) pores, verifying their continuity from the surface to the bottom of the column (approximately 700 mm).

Figure 3 is a CT image of the packed soil column with the artificially produced macropores. Compari-

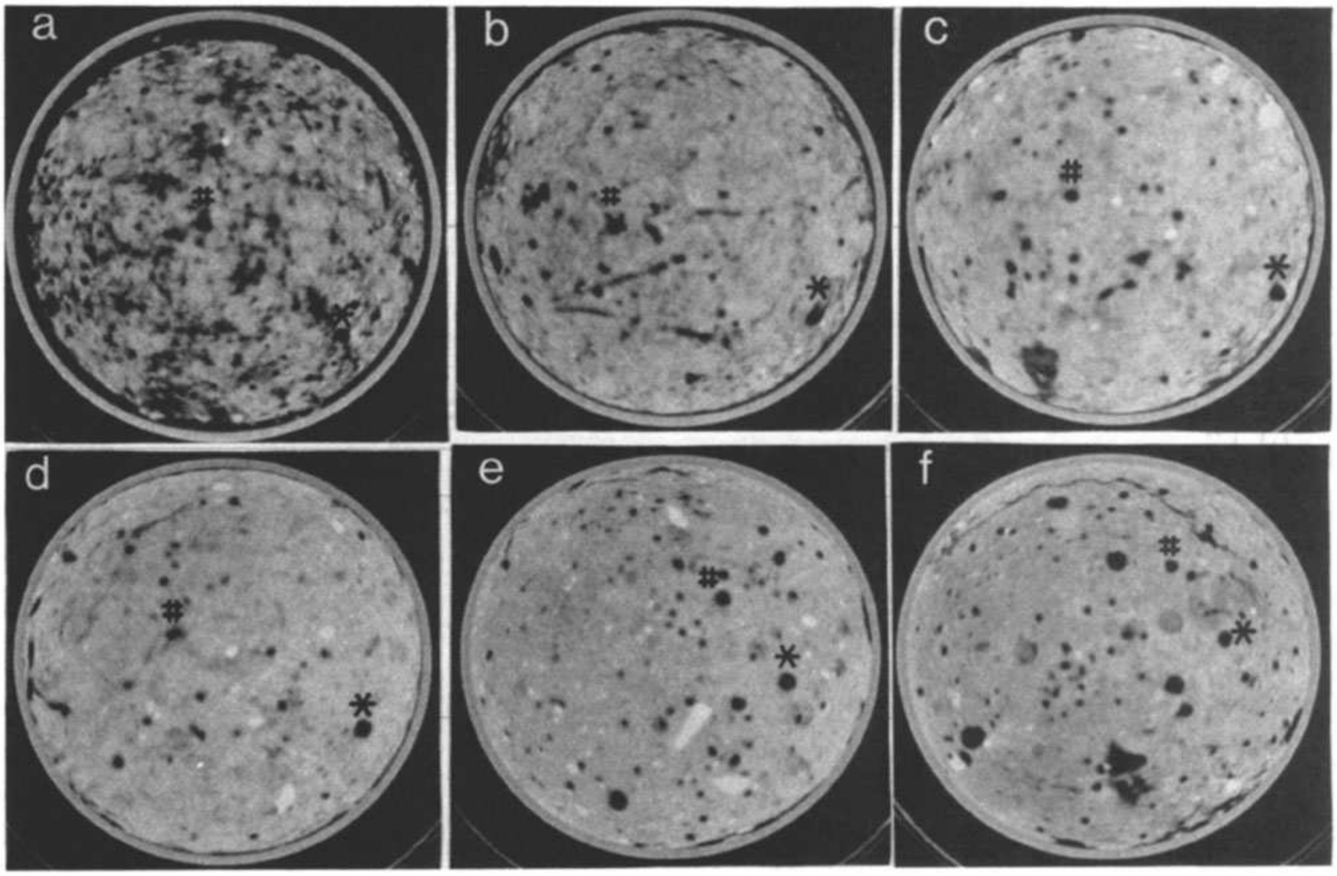

Fig. 2. CT images at indicated depths of field core. * and # = verified continuous pores. a. Image at 20 mm. b. Image at 50 mm. c. Image at 90 mm. d. Image at 130 mm. e. Image at 360 mm. f. Image at 510 mm.

son of macropores shown on this image with those created in the packed soil column demonstrated that the CT scanner accurately depicted the location and size of all the macropores. The size and type of macropores in the sample are indicated on the figure.

Four images from the field core were used as representative samples for a quantitative comparison between macropores seen on the images and those marked on the sections. In this comparison a visual check was made to determine whether macropores found on the physical section had a corresponding macropore appearing on slides made from the film acquired directly from the CT dedicated monitor. Table 1 summarizes the results. An average of 3% of macropores marked on the sections were not shown on the black and white CT images based on the total number shown on the images. The size of the macropores not shown was 3 mm or smaller. However, approximately 30% of the pores shown on the images were not marked with colored pins at the time of sectioning. Later review of section photographs revealed some apparent holes that were missed when the sections were marked. Almost all holes not marked were small, approximately 1 mm or smaller, and many were near the edge of the sections where smearing sometimes occurred during sawing of the PVC pipe.

Table 2 gives the size distribution of macropores for each scan. The numbers of macropores for size

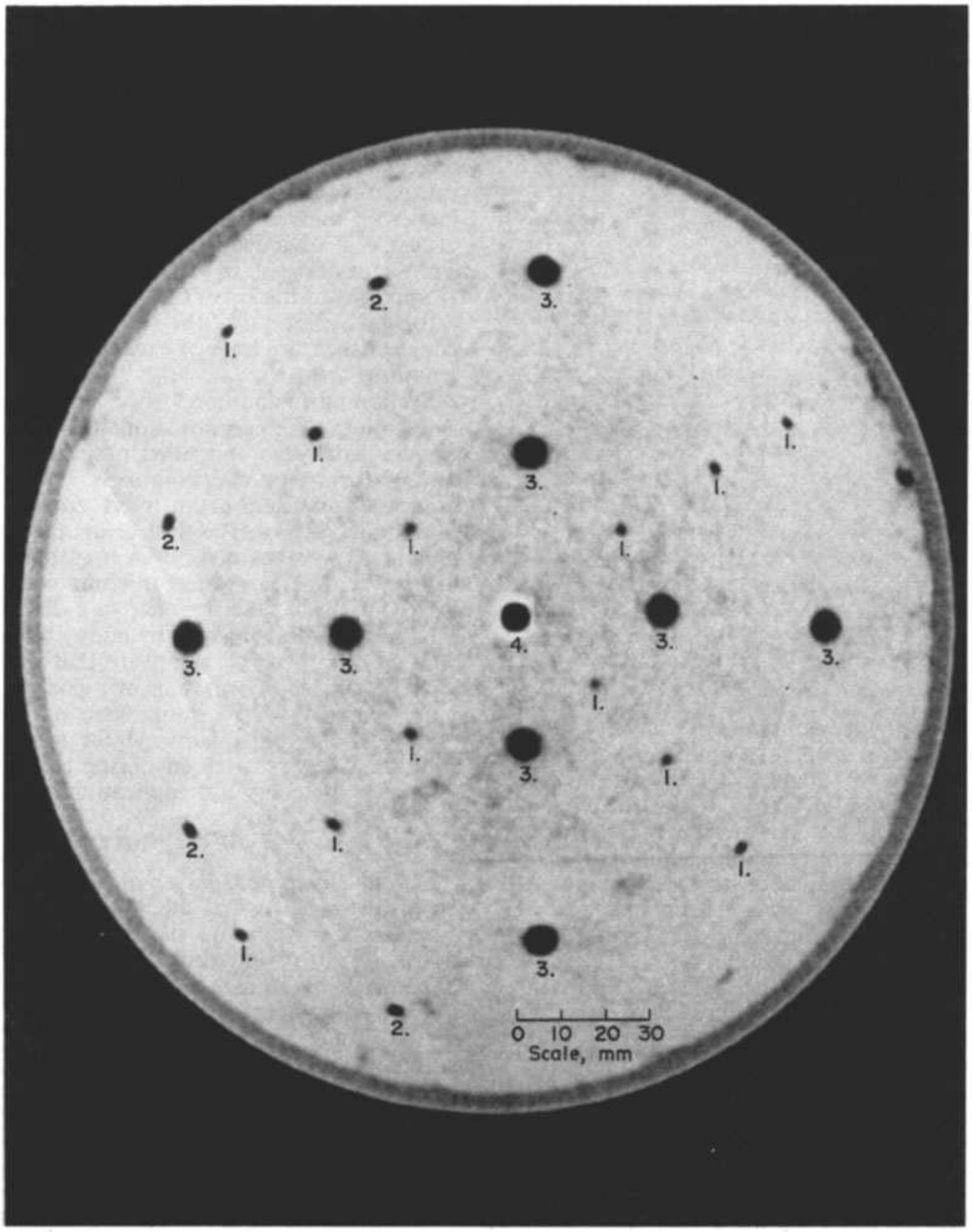

Fig. 3. CT image of packed soil column with artificial macropores. Key: 1. Hole formed by 1.57-mm diam. wire. 2. Hole formed by 2.36-mm diam. wire. 3. Plastic tube, 6.36-mm o.d., 3.16-mm i.d. 4. Glass tube, 8.13-mm o.d., 5.26-mm i.d.

Table 1. Comparisons of the number of macropores on CT images and physical sections of cores at indicated depths.

Depth, mm	Total† no. pores on image	Number pores on image, not on section	Number pores on section, not on image	Percent‡ missed in image	Percent‡ missed in section.
90	36	9	4	11	25
130	23	4	1	4	17
360	45	14	0	0	31
510	75	24	0	0	32

† Determined from black and white images acquired directly from the CT-dedicated monitor.
‡ Both of the percent missed on section and percent missed on the image are based on a total number of pores seen on the image (Column 2).

Table 2. Size distributions of macropores at indicated depth.

Depth, mm	Number of Pores† <1.0 mm	1.0–3.0 mm	3.0–5.0 mm	5.0–7.0 mm	>7.0 mm	Sum	Percent‡ air-filled
30	21	58	26	10	5	120	3.6
40	8	43	27	9	6	93	2.0
50	6	27	23	8	3	67	1.8
60	15	28	29	5	2	79	1.7
70	6	30	27	7	3	73	3.6§
80	9	15	18	8	2	52	2.2§
90	4	14	20	5	3	46	1.1
100	2	23	19	2	2	48	3.1§
110	3	23	18	1	2	47	0.9
130	3	11	15	2	1	32	0.6
150	5	11	18	2	1	37	0.7
170	3	16	19	4	3	45	1.0
190	4	17	14	4	0	39	0.7
210	2	23	15	2	1	43	0.6
230	2	22	19	3	1	46	1.0
250	4	22	15	3	1	45	1.0
270	5	24	15	2	1	47	1.0
290	4	28	10	3	1	46	2.7§
310	10	32	14	4	1	61	0.7
360	10	40	18	5	2	75	1.1
410	10	55	36	4	2	107	1.5
460	12	32	34	3	8	89	2.6§
510	7	55	29	3	8	102	2.9§
560	4	44	13	3	4	68	3.2§

† Determined from color enhanced computer displays of the images.
‡ The values in this column were calculated separately, using individual pixel values.
§ These relatively large values coincided with fauna-cavity locations—see Discussion.

ranges at each depth were estimated by visually checking images displayed on a microcomputer. Color-coded images were used for this analysis. Table 2 also reveals the changes in total porosity of macropores with depth calculated by summing the total number of pixels with an attenuation value corresponding to that of air. The total porosity in this case is considered to represent the total volume of air in pores 1 mm or larger in diameter. The reasoning for the 1-mm threshold is explained in the Discussion section.

Figure 4 presents a graph of the calculated bulk density vs. depth of scan for the field core. As pointed out earlier the density is based on the attentuation value for water and is a total or wet bulk density. The moisture content of the soil at the time of scanning was an average of about 25% on a weight basis. For a wet bulk density of 1.90 Mg m^{-3} and an assumed particle density of 2.65 Mg m^{-3}, total porosity would be 43% and dry bulk density 1.52 Mg m^{-3}. The total number of pixels used to calculate the wet bulk density for each scan was 95 093. One pixel has a volume of 0.575 mm^3 for this study, resulting in a volume per scan of 54 678 mm^3. The rapid increase in density from the

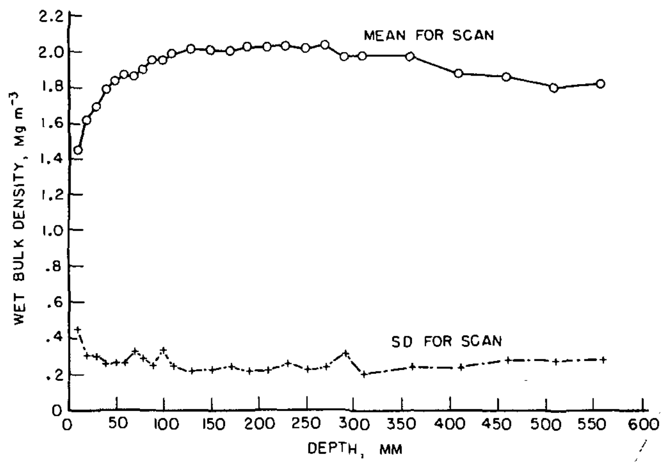

Fig. 4. Wet bulk density changes with depth-field core.

surface to the 100-mm depth is attributed to decreases of organic matter with depth in this top layer, to increases in moisture and to increases in compaction of soil particles. The grass sod covering the core had a high root mass near the surface which explains the decrease in organic matter with depth in this layer. The slight decrease in wet bulk density with depth in the bottom of the core is attributed to a drying effect through the bottom of the core, which sat for several days between the time of excavation and the time of scanning.

Sectioning of the cores revealed many soil plugs. These plugs were composed of either earthworm casts or fine sediments and filled or partially filled what were probably earthworm tunnels. The soil plugs were especially prevalent at an intermediate depth of approximately 250 to 500 mm below the surface. Some plug areas were stained, indicating that they were effectively conveying water. In some instances the CT images indicated the presence of a distinct feature at the location of a soil plug by a gray toned brightness value. Other observations were that roots often followed the sides of a pore or of a pore filled by a soil plug. Near the surface roots were stained suggesting that they may offer pathways for infiltration. Some roots may have atrophied during the 2 wk between obtaining the cores and application of the dye.

DISCUSSION

Pores 1.0 mm and larger in diameter could easily be distinguished when the images were enlarged to actual size or larger on the microcomputer. Smaller pores, down to approximately 0.5 mm, were at times distinguishable from the images but were often grayish in tone. Although the spatial resolution as determined by the scanner and image reconstruction algorithm is approximately 0.5 mm, the practical resolution is about twice this value due to boundary effects. If the boundaries of an air-filled pore 0.5 mm in diameter exactly coincided with the pixel edges, a scan image could theoretically show that pore. However, if the pore boundary falls in the middle of the pixel, the scanner will produce an attentuation value that is an average value for the materials within the pixel, for example air and soil. The probability of a 0.5-mm pore coinciding with a 0.5-mm wide pixel is small and therefore a pore must be significantly larger than 0.5 mm to have a high probability of being detected and correctly identified. A pore 1.0 mm or larger in diameter would always completely occupy at least one 0.5-mm pixel and theoretically should not escape detection.

The gray toned areas on the images may have different causes. For example, the gray toned areas seen in Fig. 2a are from roots near the surface, while the large gray area near the bottom of Fig. 2c coincides with a cavity that had been filled with paraffin. Plugs formed from earthworm casts in old macropores in the soil may also appear gray due to a lower bulk density. In addition, the gray areas could be from the boundary effect (two dimensional) or partial volume effect (three dimensional) resulting from the averaging effect of the attenuation of the x-ray beam. The pixels are actually three dimensional, being 0.536-mm by 0.536-mm by 2.0-mm long (or 8.0-mm long for the 8-mm slice width) and sometimes referred to as voxels. If a large part of the voxel is filled with air and the rest filled with a solid, the attenuation value for the voxel would be an average between that for the solid and that for the air, possibly resulting in a value corresponding to a gray shade.

The spatial resolution of the scan images may depend on the diameter of the cores. The scanner used in this study had two possible fields of view, 250 mm by 250 mm resulting in 0.536 mm pixels and 540 mm by 540 mm giving 1.0-mm pixels. These fields of view are typical of commercially available scanners. An object larger than 250 mm would require use of the larger field of view, resulting in a coarser resolution. The 203-mm diam. core was used to take advantage of the higher resolution of the smaller field of view. Larger cores would have the advantage of greater statistical significance for number of pores per area, but are more difficult to handle and result in loss of resolution.

Use of the displays on the personal computer with either the black and white or color-coded attentuation ranges, (particularly with the enlarging "zoom" feature), allowed much greater detailed examination of the scan data. Different features such as small pores and stones could be accentuated on the same image by this method. A comparison of the total number of pores at corresponding depths in Tables 1 and 2 shows that the color displays (Table 2 data) revealed a significantly larger number of pores than the photos of scan images acquired directly from the CT dedicated monitor. Practically all of the additional pores found by the color displays were very small, 1 mm^2 or less in area. The black and white images have the disadvantages of a gray scale, making detection by eye difficult. It should be noted that the ratio of the total number of pores from Table 1 to Table 2 for each depth is about the same.

Edwards et al. (1988) used a photographic/image analyzer technique to characterize macropores. When the numbers in Table 2 are converted to a square meter basis, it can be seen that the total number of pores greater than 5 mm in diameter are roughly the same for the two different techniques. However, the total number of all pores determined in this study was much less than determined by Edwards et al. (1988). The discrepancies can be explained by the resolutions of the two methods. They were able to detect pores down to 0.4 mm in diameter, and the vast majority of the pores they found were less than 1.0 mm. They attributed pores of this size to roots and not to air-filled pores. Although some pores smaller than 1.0 mm were detected by the color-coded displays in our study, most pores shown were 1.0 mm or larger. Also, the size-range distributions given in Table 2 were based on the attenuation value for air-filled pores. Pixels and therefore pores having higher attenuation values corresponding to roots were not counted for this study.

Some of the voids displayed on the CT images were large, about 20 to 40 mm across. Sectioning of the cores revealed that these large voids were ant or other insect nests or burrows, some of which were filled with eggs. The ant or other fauna burrows were not evident at the surface of the core, although a few earthworm casts could be seen at the surface after removal of the grass cover. Most of these large voids were located between 70 and 200 mm below the soil surface. The numbers of pores by size shown in Table 2 do not include these voids as the pores were counted manually from color-coded displays on the microcomputer. The air-filled porosity values do include these large cavities since they were calculated by computer, however. These voids were stained by the blue dye applied at the soil surface. They did not appear to be highly vertically continuous based on successive 10-mm spaced scans, but were connected by smaller holes to other voids and the soil surface as evidenced by the staining from the dye.

The 8-mm wide slice was used for the packed column because of the uniformity with depth of the artificial "macropores." The larger slice width has the advantage that a smoother less "noisy" image is obtained. The slice width of 2 mm was used for the field core (the minimum possible for the scanner used) to obtain as much detail as possible. In principle the x-ray beam is spread uniformly over the slice width resulting in calculated pixel values that are an average attenuation value for the entire slice width. For uniform material with vertically straight macropores, the larger slice width would be best. However, most macropores in soil appear to meander either horizontally or vertically, and as small a slice width as possible is needed to reveal detail.

The results shown in Table 1 of the comparison of the black and white scan images with the core sections showed a high correlation between the number and size of pores shown in the images with those found in the sections. At times a pore could be seen on an image that could not readily be seen on the corresponding section, but frequently probing with a pin in the section would reveal a pore that was covered by smearing or by loose soil. Very few pores were found on the sections that were not visible on the CT scan images. These differences may have been caused by uneven section surfaces, or from a cavity formed from sectioning, or vacuuming loose material such as small stones; or may be due to boundary or partial volume effects resulting in an undetected gray area as explained above. Another possible explanation of the discrepancies is the difficulty in sectioning the core exactly at the location of the scans. If the section were made a few millimeters from the actual scan, some pores may have terminated or angled off within the small distance and show on the images but not on the section, or vice versa.

Although the main objective of this paper is to report whether characterization of macropores is feasible by the CT scanning method, ancillary observations were also made. One observation is that some pores are continuous throughout the soil cores. Also, macropores exist in all the images, and there is not an abrupt change in the number of macropores from one image to an adjacent image.

Determination of the connectivity of individual pores using the CT scan images only, was difficult. Usually a pore was laterally displaced in comparing images at successive depths. Many appeared to be meandering vertical earthworm tunnels. Some earthworm-type pores near the edge of a core were at times missing in successive images, but reappeared farther down in the core. The core may have cut off a meandering tunnel near the edge in these cases. Some of the differences in the number of pores from layer to layer shown in Table 2 may be due to this, particularly for larger pores which are attributed to large earthworms such as *Lumbricus terrestris*. As reported by Lee (1985), some earthworm species form a vertical, sometimes branching tunnel. Others form predominantly horizontal tunnels that may meander randomly within certain soil layers with some openings to the surface. Still others form more or less vertical tunnels ending in roughly spherical mucous-lined chambers. Hence lateral shifts from image to image of earthworm-sized macropores is not surprising. The dye stain and sectioning process indicated that some pores were connected.

More complete assessments of the continuity and connectivity of macropores in soils using CT technology will require the reconstruction of three-dimensional images of the soil and macropore system. The methodology for reconstructing these three-dimensional images will require additional research, although a parallel methodology is already available in the medical field (Suetens et al., 1983).

In addition to the work needed in the three-dimensional reconstruction is the need to develop methodology to automate the interpretation of images produced. The scans examined in this paper were interpreted visually. An automated system would allow the rapid determination of number, size, and character of macropores, roots, stones, and other objects present in the scan image data. The algorithms and procedures developed for image analysis of photographs could probably be applied to such an automated system.

CONCLUSIONS

The CT scan process is a promising nondestructive method for the characterization of macropores in soil. Pores 1.0 mm and larger in average diameter can be readily detected from the images produced from the CT scans. The CT process is rapid and convenient when compared to a physical sectioning process, and accurately reveals the number, size and location of macropores. Problems with obtaining smear-free, uniform-plane physical cross sections make visual identification of macropores difficult. The data from the CT process can be manipulated to produce images that differentiate among materials of different densities in the soil. Wet bulk densities and air-filled porosities can be calculated for cross sections at different depths in an undisturbed soil column.

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